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[9]

SARDAR PATEL UNIVERSITY
BCA 3rd SEM EXAMINATION Feb.-2025
US03AEBCA06 : DISCRETE MATHEMATICS

Date: 24/2/2025 (Monday)

Time: 10:00 AM to 11:30 AM

Marks: 25

Q-1 Select appropriate answer.

[05]

- 1 If matrix $A = \begin{pmatrix} t-5 & 7 \\ -1 & t+3 \end{pmatrix}$ then $|A| =$ _____
 (a) $t^2 - 2t - 8$ (b) $t^2 - 2t + 8$ (c) $t^2 + 2t + 8$ (d) None
- 2 If $A = \text{diag}(2, -3, 5)$ and $B = \text{diag}(7, 4, 6)$ then AB _____
 (a) $\text{diag}(14, -12, 30)$ (b) $\text{diag}(4, -2, 0)$ (c) $\text{diag}(1, -1, 0)$ (d) None
- 3 A path in graph is _____ if all the edges are distinct.
 (a) Trail (b) Simple (c) Cycle (d) Degree
- 4 The number of edges in the complete graph K_8 is _____
 (a) 28 (b) 8 (c) 16 (d) None of these
- 5 The chromatic number of graph K_{120} is $\chi(K_{120}) =$ _____
 (a) 120 (b) 10 (c) 20 (d) 12

Q-2

Find AB where $A = \begin{bmatrix} 1 & -2 & 5 \\ -4 & -3 & 6 \\ 1 & 5 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 2 & -2 & 0 & -3 \\ 0 & 4 & -5 & 2 \\ 2 & 6 & -4 & 2 \end{bmatrix}$

[10]

Also prove that $(AB)^T = B^T A^T$.

OR

Q-2

If $A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 2 & 6 \\ 1 & 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 & 7 \\ 0 & 2 & 6 \\ 8 & 6 & 0 \end{bmatrix}$

[10]

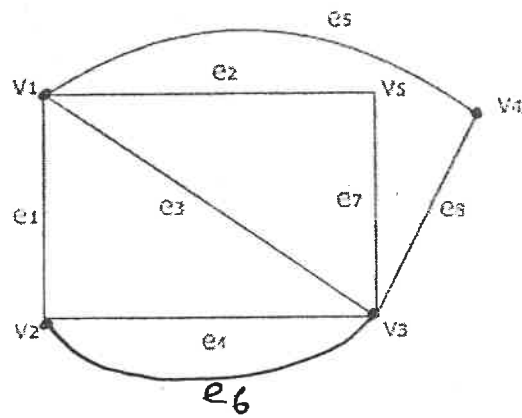
Find $A+B$, $A-2B$, $2A+3B-I$, $(2A-B+3I)^T$

Q-3 Find the Adjacency matrix and the Incidence matrix for the following graph.

[10]

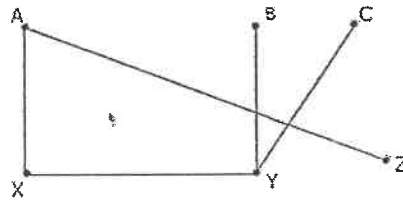
(1)

(P.T.O)



OR

- Q-3 Define bridge and cut point. Find bridge and cut points for the following graph. [10]
Also describe graph formally and find degree and parity of each vertex.



— X —

(2)